ACCREDITRACK: STREAMLINING UNIVERSITY ACCREDITATION PROCESS THROUGH A WEB-BASED SYSTEM

A Capstone Project
Presented to the Faculty
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Catanduanes State University
Virac, Catanduanes

In Partial Fulfillment of the
Requirements for the Degree
Bachelor of Science in Information Technology

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This Capstone Project entitled, "**Accreditrack: Streamlining University Accreditation process through a web-based system**" was prepared and submitted by **John Carl Sanchez, Allyza Capistrano, Carl Andrae Gonzales, and Zedric Tatad**, in partial fulfillment of the requirements for the degree of **Bachelor of Science in Information Technology.**

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EXECUTIVE SUMMARY

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(Unpublished Undergraduate Capstone Project
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This capstone project, **AccrediTrack: Streamlining University Accreditation Process Through a Web-Based System**, was undertaken for the College of Information and Communications Technology (CICT) at Catanduanes State University to modernize its predominantly manual accreditation process. Existing paper-based and e-mail workflows create fragmented procedures, excessive paperwork, and delays.

The project aimed to design and implement a secure, centralized, and user-friendly online platform that automates document uploading, validation, organization, and progress tracking. It integrates real-time dashboards and notifications to improve monitoring and communication among faculty, administrators, and quality-assurance coordinators, while ensuring data security consistent with the Data Privacy Act of 2012.

AccrediTrack consolidates accreditation records into a single repository with automated status updates, thereby reducing administrative burden, minimizing errors, and enabling timely compliance. Its prototype includes role-based user access, document management workflows, real-time progress displays, automated reminders, search and audit-trail functions, and secure storage organized by program and academic year.

Developed with PHP (Bootstrap), MySQL, HTML, CSS, and JavaScript following the Waterfall Model, the system was refined through stakeholder consultations and user testing. Evaluation using the Descriptive Survey Method confirmed that AccrediTrack

accelerates, streamlines, and enhances transparency in accreditation, offering a scalable model for other institutions pursuing digital transformation in quality assurance.

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The researchers wish to express their heartfelt gratitude to all individuals and institutions who extended invaluable assistance, guidance, and support in the completion of this capstone project.

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We sincerely thank the **College of Information and Communications Technology** and the **Quality Assurance Coordinator** of Catanduanes State University for granting access to relevant data, sharing institutional practices, and providing the necessary resources during system development and testing.

Special thanks are extended to our classmates, friends, and families for their moral support, patience, and understanding throughout the research process.

Above all, we are profoundly grateful to **Almighty God** for granting us wisdom, perseverance, and strength to complete this study.

J.C.S.

A.F.C.

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Z.T.T.

DEDICATION

This work is lovingly dedicated to our beloved families, whose unwavering support, understanding, and encouragement have been our constant source of strength and inspiration throughout the course of this study.

We also dedicate this project to our friends and adviser who guided and motivated us during the research and development of AccrediTrack. Their advice and encouragement helped us persevere and complete this work.

Above all, we offer this accomplishment to Almighty God, who granted us wisdom, patience, and resilience to overcome every challenge.

J.C.S.

A.F.C.

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TABLE OF CONTENTS

TABLE OF CONTENTS 1

LIST OF FIGURES 3

LIST OF TABLES 4

CHAPTER 1 5

Project Context 5

Purpose and Description 7

Objectives 10

Scope and Limitations 11

Definition of Terms 13

Conceptual Framework 14

CHAPTER 2 17

Synthesis 25

CHAPTER 3 27

Hardware Requirements 27

Software Requirements 28

Data Requirements 30

User Requirements……33

CHAPTER 4 34

Requirements Analysis 34

Requirements Documentation..... 37

Design of Software, Systems, Product and/or Processes..... 40

Development and Testing 41

Data Gathering Procedure..... 43

Respondents of the Study..... 45

REFERENCES 59

A. Books 59

B. Journals/Conferences 59

C. Websites..... 63

D. Other Sources..... 63

LIST OF FIGURES

Figure 1. Conceptual Framework 14

Figure 2. Use Case Diagram 40

Figure 3. Data Flow Diagram 43

Figure 4. Waterfall Model 45

LIST OF TABLES

Table 1. Hardware Requirements 28

Table 2. Software Requirements 29

Table 3. Data Requirements 31

Table 4. User Requirements 33

CHAPTER 1
INTRODUCTION

Project Context

Accreditation plays a pivotal role in the global education landscape, serving as a benchmark for quality and credibility in institutions and programs. As the world continues to evolve, so do the standards of education, making accreditation more crucial than ever. Accreditation is a formal process through which an educational institution or a specific program is evaluated against established standards of quality by an independent accrediting body. As a process, accreditation reflects the fact that in achieving recognition by the accrediting agency, the institution or program is committed to self-study and external review by one's peers in seeking not only to meet standards but to continuously seek ways in which to enhance the quality of education and training provided. Ehlers et al. (2020) indicated that accreditation could be an important tool to improve organizational quality. They agree that accreditation is an important tool to improve organizational quality, increase the motivation of staff, encourage teamwork and collaboration, and contribute to implementing changes. The higher the accreditation level of the institution, the better the program quality. Accreditation offers many benefits from the institution as this would be the basis for granting Centers of Excellence and Centers of Development among the programs in any higher education institution.

In the United States, accreditation is carried out by private, nonprofit organizations. The USA accreditation structure is decentralized and complex. There are four types of accrediting organizations in the USA. Accreditation organizations derive legitimacy from the colleges, universities, and programs that created accreditation, not the government. Accreditation of institutions and programs takes place on a cycle that may range from every few years to 10 years. The steps involve a combination of several tasks: preparation of evidence of accomplishment by the institution or the program, scrutiny of this evidence, and a site visit to the faculty and administrative peers and action by the accreditation

organization to determine accreditation status. One of the recognized accrediting agencies is Accrediting Commission of Career Schools and Colleges. This agency can accredit postsecondary, non-degree-granting institutions and degree-granting institutions in the United States. The agency provides a digital platform that enables the tracking of the accreditation process. (USA Education, 2021). Eaton, President Emeritus of CHEA, noted that as higher education institutions change and innovate, so must accreditation. Eaton asserted that the most significant shift in accreditation has been how institutions and programs are reviewed and examined (Eaton, 2020).

The institution of Catanduanes State University, in its unwavering pursuit of academic excellence and continuous institutional improvement, finds itself in the necessary position of undergoing a comprehensive accreditation process, which involves a thorough survey visit and meticulous evaluation conducted by the authorized governing body, specifically the Commission on Higher Education, working in conjunction with its designated partner agency, the Accrediting Agency for Chartered Colleges and Universities in the Philippines (AACCUP), to properly assess and validate the university's adherence to mandated quality standards in higher education.

However, the accreditation process at CatSU, particularly within the College of Information and Communications Technology (CICT), remains largely manual. It involves substantial paperwork, document sorting, and coordination between departments, faculty, and quality assurance coordinator. Klemencic and Pupinis (2021) note that traditional approaches to accreditation often result in inefficiencies due to fragmented workflows and reliance on physical documentation. Furthermore, these outdated processes are labor-intensive and error-prone, increasing the chances of misplaced files, miscommunication, and delays in submission.

The CICT department, like many academic units in developing countries, faces significant challenges in handling accreditation-related tasks. These include the lack of a unified data repository, difficulty in tracking document updates, and inadequate tools for monitoring progress. Faculty members often juggle teaching responsibilities with

administrative duties, making it difficult to meet tight accreditation deadlines (Fernández & Shaw, 2020). Additionally, without a real-time monitoring system, it becomes hard for stakeholders to stay informed about compliance status and required actions.

Coordination is another major hurdle. Different departments often work independently, making it difficult to bring everyone together under a common goal. Communication gaps can slow down the process, and it takes a lot of effort to ensure that everyone is on the same page (Singhal, 2020). Without a central system, it is hard to see where things stand or to quickly identify what still needs to be done. Resource limitations also play a big role. Many institutions in developing countries struggle with outdated infrastructure, limited access to technology, and a shortage of trained staff who understand the accreditation process. This makes it even more challenging to meet the growing expectations of students, employers, and government agencies.

In response to these challenges, the study proposes the development of AccrediTrack: Streamlining University Accreditation Process Through a Web-Based System, a project designed to modernize accreditation management. This system will automate document uploading, validation, organization, and tracking via an online platform accessible to authorized users. Its goals include reducing administrative burdens, improving document accuracy, and enhancing transparency and collaboration. AccrediTrack incorporates scalable and flexible web technologies, utilizing cloud computing for data storage and backup while implementing data encryption and user authentication for security. Through a real-time dashboard, stakeholders can monitor document completion by area, identify pending requirements, and receive automated deadline notifications. These features align with best practices in educational technology systems (Ifenthaler & Gibson, 2023).

Purpose and Description

The primary purpose of this research is to develop and implement a web-based accreditation management system called AccrediTrack, specially designed for CatSU's

College of Information and Communications Technology. The system aims to help the college handle the accreditation process in a much more organized and efficient way. Instead of relying on paper-based submissions and scattered email threads, AccrediTrack offers a single, user-friendly online platform where everything related to accreditation can be managed in one place.

Traditional methods for handling accreditation documents are known for being slow and prone to mistakes. For example, when files are managed using folders in shared drives or collected through emails, it often leads to problems like version conflicts, lost documents, and confusion over which is the latest file. These issues make it hard to keep track of changes and ensure that everyone is working with the correct information. Research shows that such manual practices can make the accreditation process inconsistent and difficult to trace, which is a major obstacle during accreditation reviews.

AccrediTrack is designed to solve these problems by storing all documents in a secure and searchable database. This means that faculty and staff no longer have to waste time searching for files or worrying about whether they have the latest version. The system ensures that every document is easy to find, up-to-date, and protected from accidental loss. By centralizing all accreditation materials, AccrediTrack helps the college avoid the common pitfalls of manual document management and supports a smoother, more reliable process.

The system integrates various modules, including a document uploading and validation module, where faculty can upload files, and the quality assurance (QA) team can review and approve them. Each file goes through validation protocols to ensure compliance with formatting and content standards. The approval and notification module automatically sends alerts to users regarding document status, pending approvals, or revision requests, improving communication loops.

One of the offered features of this system is the real-time dashboard that provides a visual snapshot of the accreditation process. It displays progress for each accreditation

area, allowing department heads and QA coordinators to assess readiness at a glance. This enables proactive decision-making and reduces the last-minute rush commonly associated with accreditation deadlines (Knight & Trowler, 2021).

Another key component is the document organization and storage module, which classifies files by accreditation area, year, and program. This reduces confusion, ensures standardized naming conventions, and enables efficient audits. To protect sensitive data, AccrediTrack enforces user-level permissions and follows data privacy guidelines as recommended by the Commission on Higher Education and the Data Privacy Act of 2012.

Using an accreditation management system like AccrediTrack brings many benefits beyond just saving time. It helps the institution maintain high standards of quality and credibility, which is important for attracting students, faculty, and funding. Accreditation is not just a one-time event-it's an ongoing process that requires constant monitoring and improvement. AccrediTrack supports this by making it easier to track progress, identify areas that need attention, and adjust strategies for improvement.

The system also encourages better teamwork and communication among faculty, staff, and administrators. By providing a single source of truth, everyone involved in the accreditation process can access the same information, reducing confusion and building a sense of shared responsibility. This collaborative environment helps create a culture of continuous improvement, which is essential for long-term success and compliance with accreditation standards.

AccrediTrack is designed to be scalable, meaning it can be used by other departments or even other institutions in the future. As more educational institutions move towards digital solutions, having a reliable and adaptable accreditation management system becomes increasingly important. Tools like AccrediTrack pave the way for better academic quality assurance and help institutions stay competitive in a rapidly changing educational landscape.

By addressing the common challenges of manual accreditation documentation, AccrediTrack offers a practical and effective solution for CatSU's College of Information and Communications Technology. It streamlines workflows, improves communication, ensures data security, and supports a culture of continuous improvement. Ultimately, AccrediTrack not only makes the accreditation process easier but also helps the institution provide better education and maintain its reputation for excellence.

Objectives

The main objective of this project is to design and develop AccrediTrack, a web-based system that streamlines the university accreditation process by automating data management, improving document organization, and providing real-time progress tracking. Specifically, it aims to:

1. To design and develop a web-based accreditation management system that integrates a centralized document repository with organized storage, easy retrieval, and validation workflows.
2. To implement real-time monitoring and communication features through dashboards and feedback mechanisms that track accreditation progress, ensure transparency, and enhance collaboration among stakeholders.
3. To ensure comprehensive data security and regulatory compliance by implementing mandatory user authentication, role-based access controls, and data protection measures to safeguard sensitive information consistent with data privacy regulations.
4. Evaluate the system's effectiveness through user testing and feedback from key stakeholders to determine its impact on streamlining accreditation processes.

Scope and Limitations

This study focuses on the development and implementation of AccrediTrack: Streamlining University Accreditation Process Through a Web-Based System for Catanduanes State University, specifically within the College of Information and Communications Technology (CICT). The system aims to address the inefficiencies of manual accreditation document management by automating submission, validation, and monitoring processes.

AccrediTrack's core features include:

1. Document Uploading and Validation: Faculty members can easily upload their files to the system. The quality assurance coordinator (QAC) of the College can then review these files and approve or request changes as needed. Each document goes through a set of checks to make sure it meets the required standards for format and content. This helps maintain consistency and quality across all submitted materials.
2. Approval and Notification System: The platform automatically sends notifications to users about the status of their documents. Whether a file is approved, needs revision, or is pending review, users receive timely alerts. This feature keeps everyone informed and reduces delays caused by miscommunication or missed emails.
3. Real-Time Dashboard: The dashboard is one of the most valuable features of AccrediTrack. It gives a clear, visual overview of the entire accreditation process. Department heads and QA coordinators can quickly see how much progress has been made in each area, which helps them make decisions and address any issues before deadlines approach. This reduces the last-minute rush and stress that often comes with accreditation cycles.
4. Document Organization and Storage: All files are organized by accreditation area, year, and program. This system of classification makes it easier to find documents and ensures that everything follows a standard

naming convention. Well-organized files also make audits and reviews much simpler and more efficient.

5. Data Security and Privacy: AccrediTrack takes data protection seriously. It uses user-level permissions to control who can view or edit certain documents, and it follows data privacy guidelines set by the Commission on Higher Education and the Data Privacy Act of 2012. This helps keep sensitive information safe and ensures compliance with regulations.

Despite its comprehensive approach, the study acknowledges the following limitations:

1. Scope Restriction to Document Management: The system focuses solely on accreditation document handling and does not cover other aspects of university accreditation, such as faculty evaluation, curriculum assessment, or student assessment systems.

2. Dependence on Internet Connectivity: As a web-based platform, the system's performance is dependent on a stable internet connection. Interruptions may affect real-time monitoring and access to uploaded documents.

3. Limited Automation in Decision-Making: While the system assists in document validation and organization, final decision-making and quality assessment still require human intervention.

4. User Adoption and Training Requirements: The effectiveness of the system depends on faculty and administrative personnel adopting the platform and undergoing the necessary training to utilize its features efficiently.

5. Scalability for Large-Scale Accreditation Processes: While designed for the CICT department, the system may require modifications to

accommodate larger institutions with more complex accreditation structures.

Definition of Terms

- **Accreditation**: The process by which an institution or program is evaluated by an external body to determine if it meets established standards of quality and integrity.
- **AccrediTrack**: A web-based system designed to automate and streamline the university accreditation process, focusing on document management, real-time tracking, and reporting.
- **Automation**: The use of technology to perform tasks with minimal human intervention, aiming to increase efficiency and reduce errors (Cruz & Ramos, 2023).
- **Centralized Document Management**: A system where all accreditation-related documents are stored and organized in a single, accessible location, facilitating easy retrieval and management.
- **Cloud Computing**: The delivery of computing services including servers, storage, databases, networking, software, analytics, and intelligence over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.
- **Commission on Higher Education (CHED)**: Is responsible for managing and formulating policies related to tertiary education.
- **Dashboard**: A visual interface that provides real-time summaries and key performance indicators, enabling users to monitor progress and make informed decisions.
- **Data Collection**: The process of gathering and measuring information on targeted variables in an established systematic fashion, which then enables one to answer relevant questions and evaluate outcomes.

- **Database Management**: The process of storing, organizing, and managing data within a database system to ensure data integrity, security, and accessibility.
- **Digital Transformation**: The process of using digital technologies to create new or modify existing business processes, culture, and customer experiences to meet changing business and market requirements.
- **Document Validation**: The process of checking documents to ensure they meet specified criteria and standards, confirming their accuracy and completeness.
- **Higher Education Institution (HEI)**: An organization that offers undergraduate, graduate, and research programs, typically awarding degrees such as bachelor's, master's, and doctoral degrees.
- **Quality Assurance**: A systematic review of educational provision to maintain and improve quality, equity and efficiency.
- **Real-Time Monitoring**: The continuous tracking and display of data or progress updates as they occur, providing immediate insights and feedback.
- **Web-Based System**: A software application that is accessed through a web browser over the internet, allowing for remote access and collaboration.

Conceptual Framework

The conceptual framework includes the following diagrams, which describe the required inputs and outline how the process functions to produce the intended output.

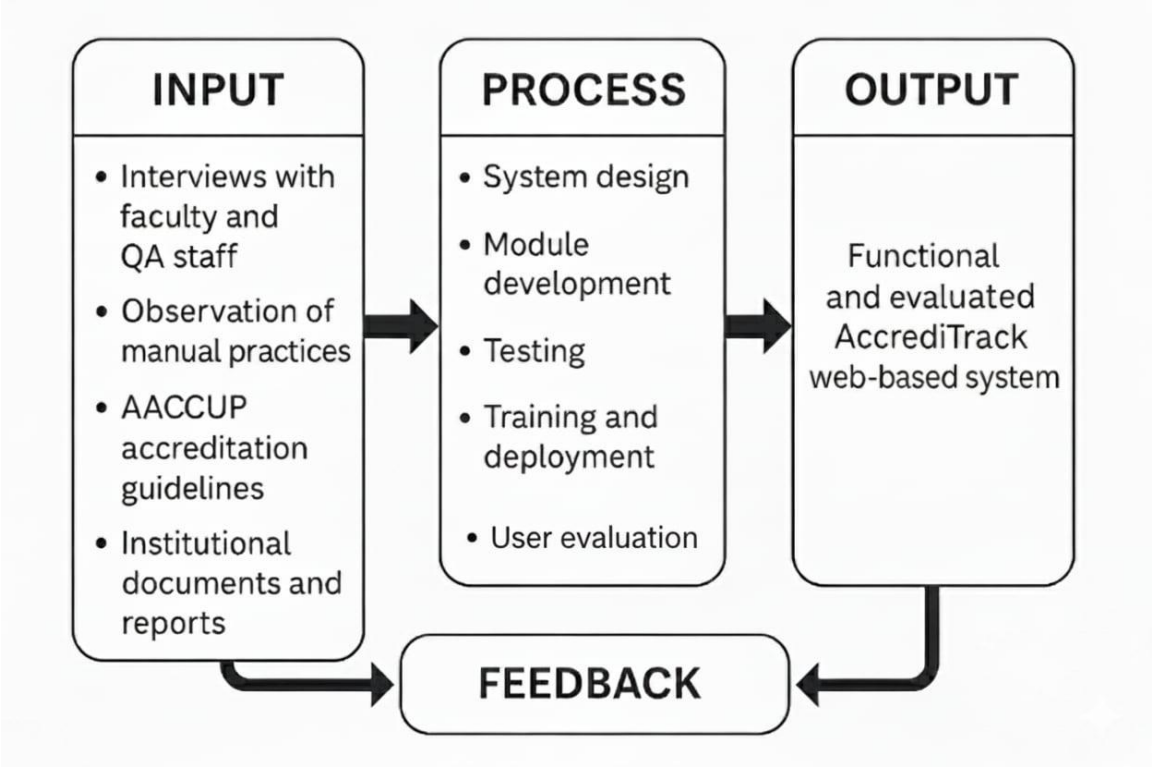


Figure 1. Conceptual Framework

Figure 1. Conceptual Framework presents the development of the AccrediTrack web-based system using an Input–Process–Output (IPO) model with an integrated feedback loop.

In the input phase, relevant information is gathered from institutional stakeholders—including faculty and the Quality Assurance Coordinator—through interviews, observation of current manual practices, review of AACUP accreditation guidelines, and analysis of institutional documents and reports. These inputs inform the system requirements and design specifications. The process phase involves translating these inputs into action through system design, module development, testing, and user training leading to deployment. Key modules include document upload and validation, real-time progress dashboards, and automated notifications for revisions or errors. The output phase yields a functional and evaluated AccrediTrack system that streamlines

accreditation workflows, improves collaboration, and enhances transparency and accountability across the institution. A built-in feedback mechanism connects the output back to the input stage, enabling continuous refinement of system features based on user experiences and evaluation results, thus ensuring sustainability and responsiveness to evolving accreditation needs.

CHAPTER 2
REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter discusses the reviews of related research and techniques relevant to the proposed system, AccrediTrack: Streamlining University Accreditation Process Through a Web-Based System. It includes studies on accreditation, educational technologies, document management, evaluation methods, real-time data analytics, and existing system implementations. The literature is presented thematically, aligning with the project's development and evaluation objectives. These thematic areas provide insights into the current practices, challenges, and innovations that inform and support the system's development.

Accreditation Processes and Challenges in Higher Education

Accreditation serves as a critical quality assurance mechanism in higher education, aimed at aligning academic programs with predefined standards. Duarte and Vardasca (2023) regard accreditation as an essential verification tool that reinforces institutional commitment to continuous improvement. Similarly, Kumar, Shukla, and Passey (2021) emphasize its role in enhancing institutional performance and academic excellence. However, as higher education institutions operate as open systems, achieving successful accreditation outcomes hinges on effective collaboration among stakeholders—particularly faculty members (Muljana, Nissenson, & Luo, 2020).

In the Philippine context, studies reveal significant burdens placed on faculty during accreditation cycles. Panela and Deniega (2021) report that educators often face excessive workloads related to preparing and compiling voluminous documentation, which detracts from their core responsibilities and causes stress. Manual retrieval of supporting documents from various offices remains one of the most labor-intensive aspects of the process. At institutions like Cebu Technological University, reliance on printed

documentation exacerbates inefficiencies, making data collection, presentation, and organization time-consuming and disorganized.

Romanowski (2020) argued that the accreditation process is shaped by unequal power dynamics, allowing accrediting bodies to exert significant influence over universities. These agencies can make extensive demands, which institutions often feel compelled to meet in order to achieve or retain their accreditation status. Despite limited resources, universities are frequently guided and controlled by these agencies due to the perceived necessity of accreditation—often reallocating funds from other institutional priorities to fulfill accreditation requirements.

Accreditation's importance extends beyond compliance; it affects the reputation and global competitiveness of higher education institutions. According to Kumar (2020), more than 80% of students consider accreditation and institutional ranking when selecting academic programs. Ravikumar et al. (2021) add that quality assurance fosters innovation and improved learning outcomes. In Asia, while national accreditation bodies are widespread, international accreditation remains more desirable due to broader recognition and facilitation of student mobility, academic exchange, and credit transfers.

Recent discussions emphasize the importance of establishing structured and strategic planning in the accreditation processes of higher education institutions. Semper (2020) underscored the need for a consistent and systematic approach in gathering and managing information relevant to accreditation. Ramos et al. (2020) emphasized that integrating ISO standards into university operations leads to more efficient use of institutional resources and improved stakeholder satisfaction. Their research highlighted academic audits as essential tools for monitoring accreditation-related activities and institutional improvement.

As the education sector continues to evolve in response to societal changes, the processes ensuring its quality and relevance must also adapt. It is essential for all stakeholders—including policymakers, educators, students, and their families—to grasp

the mechanisms of accreditation and their significance in shaping educational institutions (Duarte & Vardasca, 2023).

Based on the study, it can be concluded that while accreditation is essential for maintaining quality standards, enhancing institutional reputation, and promoting global competitiveness in higher education, it also imposes substantial challenges, particularly on faculty members who bear the weight of documentation and compliance tasks. The manual and labor-intensive nature of accreditation processes in many Philippine institutions highlights the need for more efficient, technology-driven solutions. Also, the influence of accrediting bodies and the pressure to meet their demands often leads institutions to reallocate resources, sometimes at the expense of other priorities.

To address these issues, the study emphasizes the importance of structured planning, integration of ISO standards, and academic audits as tools for improving institutional efficiency and stakeholder satisfaction. Ultimately, a collaborative, well-managed accreditation process is key to fostering continuous improvement and sustaining the relevance of higher education in a rapidly evolving society.

Digital Systems for Accreditation Management and Documentation

To address the burdens of traditional accreditation methods, the integration of technology—especially document and data management systems—has become a key focus in modern educational administration. Electronic Document Management Systems (EDMS) simplify accreditation workflows by offering centralized and organized digital repositories. Estrera, Fajardo, and Sermona (2022) highlight the ability of EDMS to reduce reliance on paper, improving operational efficiency, sustainability, and cost-effectiveness. Ramey (2020) supports this view, stating that educational technologies not only streamline access to information but also improve user engagement.

Manish and Zainab (2024) observed a significant transformation in the accreditation process, noting a shift from manual, error-prone procedures to modern, automated systems. Platforms like LearnQoch have introduced centralized data management and automated reporting, significantly enhancing efficiency and accuracy in handling accreditation tasks. Similarly, Alechko (2025) highlighted the inefficiencies of manually generating crucial documents such as accreditation decisions and renewals. Automation in these processes minimizes errors and improves institutional performance by reducing administrative workload and streamlining communication.

Rosiska (2020) notes that digital documentation significantly accelerates retrieval and improves organizational clarity during accreditation presentations. Cloud-based storage systems represent an evolution in this domain by offering scalable, secure, and collaborative access to documents across institutional departments. However, despite these advancements, their application remains limited in some Philippine institutions. For instance, Cebu Technological University still heavily relies on traditional paper-based documentation methods, illustrating a clear gap that systems like *AccrediTrack* aim to address.

Several institutions have experimented with partial solutions. Kommey et al. (2022) introduced document tracking via email alerts but fell short of offering full automation. Similarly, DocTract (Munro & Baird, 2018) enhances communication during accreditation with automated alerts—a feature mirrored in AccrediTrack to facilitate real-time collaboration.

Additionally, Chawla et al. (2020) observed that automation through Online Management Systems (OMS) significantly reduced time spent on manual tasks such as document handling, student applications, and course registration—ultimately benefiting both administrators and students.

The findings of the study suggest that the integration of digital systems in accreditation management has significantly transformed traditional practices by

streamlining documentation, reducing administrative burden, and improving overall institutional efficiency. Technologies such as Electronic Document Management Systems (EDMS), cloud-based platforms, and Online Management Systems (OMS) have proven effective in centralizing data, automating reporting, and enhancing collaboration among stakeholders. While global trends show a shift toward digital and automated processes, the study also highlights that many Philippine institutions still lag, relying on outdated, paper-based methods. This underscores the need for broader adoption of advanced digital solutions like AccrediTrack to modernize accreditation workflows and meet the growing demands of quality assurance in higher education.

Real-Time Data Analytics and Dashboard Design for Evaluation

Modern systems increasingly rely on real-time data analytics to support decision-making and improve process outcomes. In the context of accreditation and administrative systems, real-time analytics can facilitate faster data retrieval, monitoring, and reporting. Miller and Dubrawski (2020) describe how real-time surveillance supports proactive actions, particularly in operational environments. Predictive analytics further strengthens this approach by identifying trends or anomalies that indicate emerging issues, allowing institutions to take preventive measures before problems escalate (Maddireddy & Maddireddy, 2022).

A critical component of real-time evaluation systems is the dashboard interface. Ludlow et al. (2021) argue that effective dashboards must prioritize clarity, simplicity, and customization. Visual tools such as gauges, charts, and graphs enhance user comprehension of live data. Customizable dashboards cater to different user roles—managers may require high-level summaries, while staff benefit from task-specific views. Additionally, dashboards with alert systems triggered by thresholds or algorithms ensure timely responses to issues (Bachechi, Po, & Rollo, 2022).

Effective dashboard design is crucial for using the benefits of real-time data analytics in safety monitoring and administrative evaluations. A well-designed dashboard should be clear, concise, and actionable, enabling users to make informed decisions quickly. Simplicity in visual elements—charts, gauges, and color-coded indicators—helps avoid information overload and directs focus to key metrics. Real-time visualization, supported by data integration, is essential to detect anomalies or issues as they arise. Moreover, alert systems and role-based customization further improve the responsiveness and relevance of dashboard interfaces.

In Indonesia, Hasanati et al. integrated a dashboard system into their Academic Information System, allowing academic executives to periodically evaluate internal program status in line with accreditation standards. Similarly, Gunawan et al. developed a dashboard system at Universitas Pendidikan Ganesha using the Extreme Programming model, facilitating accreditation monitoring and addressing the lack of data integration and real-time oversight.

Based on the study, it is evident that real-time data analytics and well-designed dashboard systems play a pivotal role in enhancing the efficiency and responsiveness of accreditation and administrative evaluations in higher education. By enabling instant data retrieval, monitoring, and predictive insights, institutions can make timely, evidence-based decisions and address potential issues before they escalate. Effective dashboard design—characterized by simplicity, clarity, and role-based customization—supports users at all levels in navigating complex information with ease. The inclusion of alert mechanisms and visual tools such as charts and color-coded indicators further enhances situational awareness and prompts action. Examples from Indonesia illustrate how real-time dashboards, when integrated into academic information systems, improve alignment with accreditation standards and foster continuous institutional improvement.

Real-Time and Predictive Data Quality Management

Data Quality Management (DQM) is essential in accreditation and institutional systems where accuracy and reliability are non-negotiable. Traditional DQM methods relied heavily on batch processing and manual reviews (Acquisti et al.; LaValle et al.; Davenport & Ronanki, 2020). In contrast, real-time monitoring enables continuous evaluation of data consistency, completeness, and integrity, facilitating prompt corrective actions (Siiman et al., 2023; Wang et al., 2021).

Predictive analytics allows for proactive DQM by forecasting data issues based on historical trends (Rangineni & Bhardwaj, 2024). These insights help institutions preemptively correct problems before they impact evaluations or decision-making. Machine learning enhances this capability by detecting patterns invisible to human analysts (L. W., 2020). Esther et al. (2023) and Antonio (2023) note that such integrated solutions are particularly vital in high-stakes sectors, including education, where data quality influences accreditation outcomes, compliance, and performance tracking.

Based on the gathered literature, it is apparent that real-time and predictive Data Quality Management (DQM) have become crucial components in ensuring the accuracy, reliability, and integrity of institutional data, especially in the context of accreditation and compliance. Unlike traditional batch processing methods, real-time DQM allows for continuous monitoring and immediate correction of data issues, thereby minimizing risks and enhancing responsiveness. Predictive analytics, supported by machine learning, enables institutions to identify and address potential data quality problems before they affect evaluations or decision-making processes. These advanced approaches are particularly important in education, where data plays a pivotal role in performance assessment, regulatory compliance, and maintaining accreditation standards.

Existing Accreditation Systems and Development Models

Numerous systems have been developed to support the accreditation process in higher education. The Dokumen Akreditasi Online (DAKON) system in Indonesia was developed using the Software Development Life Cycle (Waterfall model), with modules for organizing and storing accreditation documents. It achieved a 75.27% user satisfaction rate in pilot testing (Finalita et al., 2020). Similarly, Mapúa University employed the Agile Development Lifecycle to build a digital monitoring system, allowing for iterative improvements and broad compatibility across browsers, validated through tools like the Requirements Traceability Matrix (Chua et al., 2023). AccrediTrack adopts a similar Agile approach to ensure responsiveness to user feedback and cross-platform optimization.

In the Philippines, Barbosa et al. evaluated the Accreditation Online Management System (AOMS), which received a System Usability Scale (SUS) score of 74, indicating above-average usability. Stakeholders praised its user-friendly interface and collaborative features, reinforcing the value of systems that involve users actively in the accreditation process.

Web-based platforms also contribute to greater stakeholder engagement and transparency. According to Wardhani and Suhdi (2020), university websites serve as vital tools for disseminating information to both primary (academic community) and secondary (state, general public, private companies, etc.) stakeholders.

An overview of the findings reveals that the development and implementation of digital accreditation systems have significantly improved the efficiency, usability, and stakeholder engagement in higher education quality assurance. Various development models, such as the Waterfall and Agile methodologies, have been successfully applied to create systems tailored to institutional needs. For instance, Indonesia's DAKON system and Mapúa University's Agile-based platform demonstrate how structured and iterative development approaches can yield high user satisfaction and operational effectiveness. In the Philippines, the Accreditation Online Management System (AOMS) has shown strong usability and collaborative capabilities, underscoring the importance of user-centered

design. Furthermore, web-based platforms play a critical role in promoting transparency and engaging a wider range of stakeholders, enhancing institutional credibility and public trust in the accreditation process.

Synthesis

The reviewed literature identifies accreditation as a cornerstone of quality assurance in higher education, reinforcing institutional performance, credibility, and competitiveness. However, the process is often marred by documentation challenges, administrative burdens, and inefficient systems. Studies from both international and local contexts stress the urgent need for technological interventions to streamline and digitize accreditation tasks. Electronic document management, cloud-based systems, and real-time analytics offer tangible solutions that improve efficiency, transparency, and accuracy.

Furthermore, the use of dashboards, predictive analytics, and data quality frameworks demonstrates how evaluation mechanisms can evolve from reactive to proactive. Well-designed dashboards—featuring real-time visuals, role-based customization, and automated alerts—maximize insight and minimize delays. A few software systems have attempted to automate portions of this process, yet none have fully addressed the end-to-end automation of accreditation. Systems such as EAMIS, DAKON, AOMS, and institutional dashboards in Indonesia and Saudi Arabia show the potential and feasibility of technology-driven accreditation—but adoption remains inconsistent.

Digital solutions like EDMS, OMS, real-time analytics, and dashboards offer practical improvements by streamlining tasks, improving data accuracy, and supporting timely decisions. Predictive tools and machine learning further enhance data quality management. Development models such as Waterfall model have been effectively used to build user-friendly, adaptable systems that engage stakeholders and promote transparency.

This research, through the development of AccrediTrack, aims to bridge these gaps by consolidating best practices from these systems into a unified, web-based platform specifically tailored for higher education institutions in the Philippines. Through features such as centralized document management, real-time analytics, stakeholder collaboration, and automated alerts, AccrediTrack aspires to transform accreditation from a manual obligation into a streamlined, intelligent, and sustainable process.

CHAPTER 3
TECHNICAL BACKGROUND

This chapter outlines the technical foundation necessary for the development and implementation of AccrediTrack, a web-based system aimed at improving the accreditation process of the College of Information and Communications Technology (CICT) at Catanduanes State University. It describes the system's technical components, including the hardware and software requirements, the data requirements, and the specific features and modules based on user roles and project goals. These technical elements serve as the blueprint to ensure the system is fully functional, user-friendly, and aligned with institutional needs.

Hardware Requirements

Table 1. Hardware Requirements

Hardware Requirement	Specification	Description
Server Computer	Minimum: Intel Corei5 or equivalent, 8GB RAM, 512GB SSD	Hosts the web-based system, handles data processing and storage.
Client Computers	Minimum: Intel Core i3 or equivalent, 4GB RAM, 100GB HDD	End-user computers used by administrators, staff, and other stakeholders to access the system via a web browser.
Network Infrastructure	Ethernet or Wi-Fi	Reliable network connectivity is essential for accessing the system.

		Bandwidth requirements will depend on the number of concurrent users.

In Table 1, the researchers have identified the essential hardware requirements needed for the development and deployment of the AccrediTrack system. The server computer must meet at least a minimum specification of Intel Core i5, 8GB RAM, and a 512GB SSD to ensure it can handle the system's processing demands and data storage. Client computers, which will be used by administrators, QA Coordinator (QAC), and faculty, must have at least an Intel Core i3 processor, 4GB RAM, and 100GB HDD to efficiently access the system through a web browser. Reliable network infrastructure, either via Ethernet or Wi-Fi, is also necessary to support uninterrupted system access, especially during peak usage. These requirements are set to guarantee system performance, responsiveness, and accessibility across various user roles and locations.

Software Requirements

Table 2. Software Requirements

Software Requirement	Specification	Description
Frontend	HTML5, CSS3, JavaScript Framework: Bootstrap/React	Ensures a responsive, user-friendly interface.
Backend	PHP (Laravel) Python (Django optional)	Handles business logic and server-side operations.
Software Requirement	**Specification**	**Description**

Database	MySQL/PostgreSQL	Stores accreditation documents, user data, and logs.
Web Server	Apache/Nginx	Hosts the application and manages HTTP requests.
Operating System	Windows/Linux	Compatible with both server and client environments.
Browser Support	Chrome, Firefox, Edge (latest versions)	Ensures cross-browser functionality.

In Table 2, the researchers have outlined the software requirements needed during the development and deployment of the system. The frontend technologies, including HTML5, CSS3, and JavaScript with Bootstrap or React, are intended to ensure a responsive and user-friendly interface. For backend development, PHP with Laravel is used to manage the core logic and processes of the system. MySQL or PostgreSQL serves as the database for storing accreditation-related data and user information. Apache or Nginx is selected as the web server to handle and deliver client requests efficiently. The system is compatible with both Windows and Linux operating systems, allowing flexibility in both development and deployment. Additionally, the system supports major web browsers such as Chrome, Firefox, and Edge to ensure accessibility and smooth performance across different platforms.

Data Requirements

Table 3. Data Requirements

Construct/ Variables	**Source of Data**	**Data Gathering Technique**	**Instrument**	**Data Processing**
Current accreditation process and challenges	Faculty members, QAC	Interviews, Focus groups	Interview guides, Audio recording device, Note-taking	Transcribed, coded, and thematically analyzed for recurring issues
Accreditation Documents	Faculty Members, QAC	Interviews, Document Review	Document checklist, Interview guides	Sorted, categorized by area, and uploaded to system modules
System requirements and preferences	QAC, Faculty Members	Focus group, Document analysis	Interview guides, Discussion guides	Mapped to system features and prioritized by frequency

Document flow and validation process	QAC, Document handlers	Document analysis	Existing forms and process flowcharts	Converted into system workflows for automation
User Information	Registrar, QAC, Faculty Members	Interviews	Online forms, Registration forms	Digitized and entered into the system database for access control

In Table 3, the researchers identified the key data requirements essential for designing and developing AccrediTrack. Information about the current accreditation process and challenges was gathered through interviews and focus group discussions with faculty and quality assurance coordinator (QAC). This qualitative data was transcribed and analyzed to identify common pain points that the system aims to address. Accreditation documents were obtained via interviews and document review, and then sorted and organized by accreditation area within the system for easy tracking and validation.

System requirements and user preferences were also collected through discussions and document analysis to ensure that the design aligns with stakeholder expectations. These inputs were translated into specific system features and prioritized during development. The existing document flow and validation process were analyzed from current forms and flowcharts, which were then used to model and automate workflows in the system. Lastly, user information was acquired from the registrar and formatted into a digital structure for account registration and role-based access within the system. This structured approach to

data gathering and processing ensured that the system design was both practical and aligned with real institutional needs.

User Requirements

Table 4. User Requirements

Module	Function	Description	User Role
Dashboard	System monitoring	Displays real-time accreditation progress and summary of accreditation status	QAC, Faculty
Document Management	File processing	Handles document uploads (Faculty) and review/approval (Admin/QA)	QAC, Faculty
User Management	Access Control	Manages user accounts and permissions to ensure secure access to sensitive accreditation data and system features.	Admin/QAC
Notifications	Task alerts	Sends automated reminders for submissions and approvals	QAC, Faculty
Audit Logs	Activity tracking	Records all system actions for compliance and accountability	Admin/QAC

In Table 4, the researchers have outlined the key user requirements for AccrediTrack, detailing essential modules and their functions. The system employs a streamlined two-role structure: Faculty members who submit documents and Admin/QAC who oversee the entire accreditation process. The Admin/QAC role combines traditional administrative functions with quality assurance tasks, enabling comprehensive management through a unified interface. Faculty users interact primarily with the document submission features, while Admin/QAC users access all monitoring, approval, and reporting tools. This streamlined approach reduces role complexity while maintaining all necessary functionality for efficient accreditation management. The system maintains strict role separation between regular faculty and the privileged Admin/QAC account to ensure data security and process integrity.

CHAPTER 4
DESIGN AND METHODOLOGY

This chapter discusses the project technical description, system features, and methodology used for the project. Understanding the blueprint and the approach is crucial to appreciating how AccrediTrack aims to revolutionize the accreditation landscape. Therefore, this chapter will first articulate the technical underpinnings of the proposed web-based system, detailing its key features and functionalities designed to streamline the accreditation process. Following this, a comprehensive overview of the research methodology employed to guide the development, implementation, and evaluation of AccrediTrack will be presented. This will encompass the strategies for requirements gathering, system design, development approach, and the methods for assessing the system's effectiveness in achieving its objectives.

Requirements Analysis

The requirements analysis stage was conducted to determine the exact needs and expectations of the intended users of AccrediTrack. The researchers engaged in consultations with faculty members and quality assurance coordinators (QAC) of the college. In addition, existing accreditation workflows at CatSU were reviewed, and recent literature related to digital accreditation systems was examined to align the system’s functionality with academic standards and best practices.

Consultation with Stakeholders: In-depth interviews and discussions were conducted with individuals directly involved in the accreditation process. This included:

- **Quality Assurance Coordinator:** To identify their needs for preparing accreditation documents, tracking progress, communicating with the accreditation body, and managing internal workflows related to accreditation. Their perspectives

on user-friendliness and efficiency were critical in defining the user interface and overall system usability.

- **Faculty:** To understand the technical constraints, security requirements, and integration needs of the proposed web-based system within the existing technological infrastructure of the institutions and the accreditation body.

Review of Related Literature: Existing research on accreditation processes, web-based system development, and workflow automation was reviewed. This included studying best practices in accreditation management, analyzing the features of similar systems in other domains, and understanding the theoretical frameworks underpinning efficient information management and process optimization. This review provided a broader context for the project and helped identify potential challenges and innovative solutions.

Analysis of Existing Systems and Processes: The current manual and semi-automated processes used for accreditation were meticulously analyzed. This involved observing the flow of information, identifying bottlenecks, understanding the documentation requirements at each stage, and assessing the efficiency and effectiveness of the existing tools and methods. This analysis highlighted the specific areas where a web-based system could offer significant improvements in terms of speed, accuracy, and transparency.

Functional Requirements

The functional requirements define what specific services and features the system must deliver. These were identified as follows:

- **User Registration and Login:** Users must be able to register, authenticate, and access the system based on assigned roles (e.g., Admin, Staff).

- **Document Uploading:** Users should be able to upload accreditation-related documents into organized folders by program, area, and academic year.
- **Document Validation Workflow:** Quality Assurance Coordinator (QAC) must be able to review submitted documents, provide feedback, and approve or reject files.
- **Dashboard Display:** The system should provide a real-time visual overview of accreditation progress across different areas and programs.
- **Notification System:** Users should receive automated email or in-system alerts regarding submission deadlines, document status, and required revisions.
- **Search and Filter:** The system should allow searching and filtering of documents based on metadata such as area, year, status, and user.
- **Audit Trail:** System should log all actions performed by users for monitoring and accountability.

Non-Functional Requirements

These define the system's performance expectations beyond its core functionality:

- **Usability:** The system interface must be user-friendly and require minimal training for faculty and staff.
- **Security:** User authentication, role-based access control, and data encryption must be implemented to protect sensitive accreditation files.
- **Performance:** The system should handle multiple users accessing and uploading documents concurrently without performance degradation.
- **Scalability:** The system architecture must allow expansion to accommodate more programs or even different colleges within the university.

- **Reliability:** The system must be consistently available, with minimal downtime, especially during critical accreditation periods.
- **Compatibility:** The system must work across all modern browsers and support mobile responsiveness when necessary.

Requirements Documentation

To maintain clarity, traceability, and stakeholder alignment, the researchers developed a structured documentation of the requirements collected during the analysis phase. This documentation served as a guide during the design and implementation phases and also provided a reference for user acceptance testing.

Use Case Diagram

A use case diagram was created to visually represent the interactions between the system and its different user roles. It identifies actors (Admin/QAC, Faculty/Staff) and the actions they perform, such as uploading files, validating documents, generating reports, and viewing dashboards.

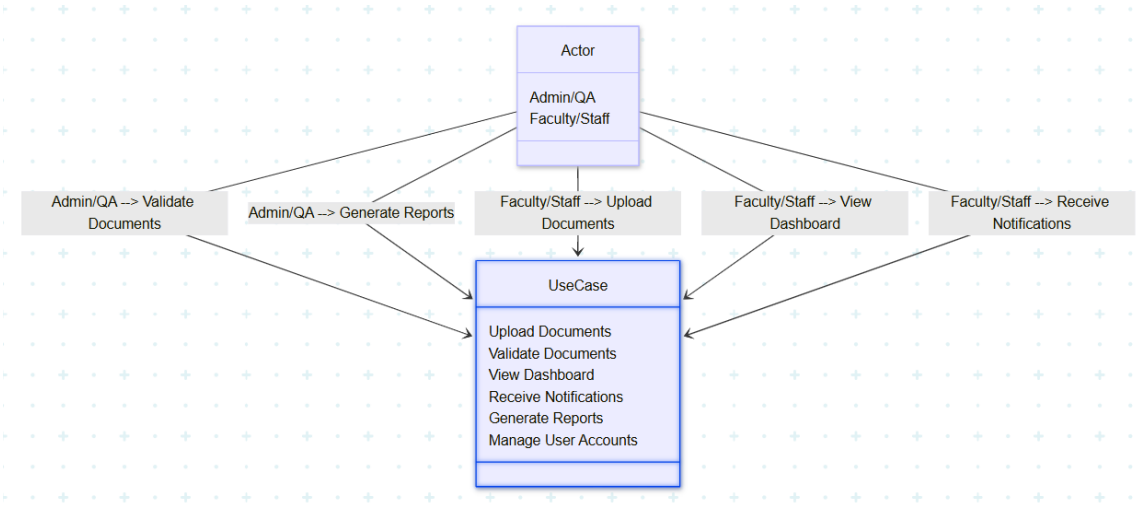


Figure 1. Use Case Diagram for AccrediTrack

User Stories

User stories were written in a simple, structured format to describe features from the user's perspective:

- **As a faculty member**, I want to upload accreditation documents so that I can fulfill my program's compliance requirements.
- **As a Quality Assurance Coordinator (QAC)**, I want to validate and provide feedback on uploaded documents so that I can ensure all materials meet accreditation standards.
- **As an admin**, I want to monitor overall progress via a dashboard so that I can coordinate efforts efficiently and prepare for survey visits.

Constraints

The development and deployment of AccrediTrack are subject to several constraints that may impact its scope and functionality. First, the system must strictly comply with institutional policies and regulatory guidelines, particularly those set by

CHED and the Data Privacy Act of 2012, to ensure the confidentiality and protection of sensitive academic data. Additionally, the system is designed to operate within a web-based environment, which requires a stable internet connection, thereby limiting functionality in offline or low-connectivity scenarios. The project is also constrained by time and resource availability, as it must be completed within the academic calendar and using only the tools and infrastructure accessible to the development team. Furthermore, the system is initially tailored for the College of Information and Communications Technology (CICT), which may require future modifications for adoption in other departments or campuses.

Assumptions

Several assumptions were made during the planning and development of AccrediTrack. It is assumed that all users, including faculty members and quality assurance staff, possess basic computer literacy and are familiar with navigating web browsers. The researchers also assume that stakeholders will actively participate during data gathering, feedback, and testing phases, ensuring the system addresses real user needs. Another assumption is that the university will provide sufficient server infrastructure or cloud hosting support to deploy the system securely and reliably. It is also assumed that the accreditation process will remain consistent in structure, allowing the system's document workflows and validation modules to remain effective across multiple accreditation cycles.

Dependencies

The success of AccrediTrack depends on several external and internal factors. One primary dependency is the availability of updated institutional accreditation guidelines, program documents, and supporting files from faculty and QAC. Without these, the system cannot fully populate or validate accreditation areas. The project also relies on continuous collaboration with stakeholders, including timely responses from QAC and administrators for testing and approval. Moreover, system deployment depends on the support of the

university's IT team, particularly in configuring servers, maintaining security protocols, and providing user training. Lastly, user feedback is a crucial dependency for system refinement, ensuring that improvements are aligned with actual workflow needs and accreditation procedures.

Design of Software, Systems, Product and/or Processes

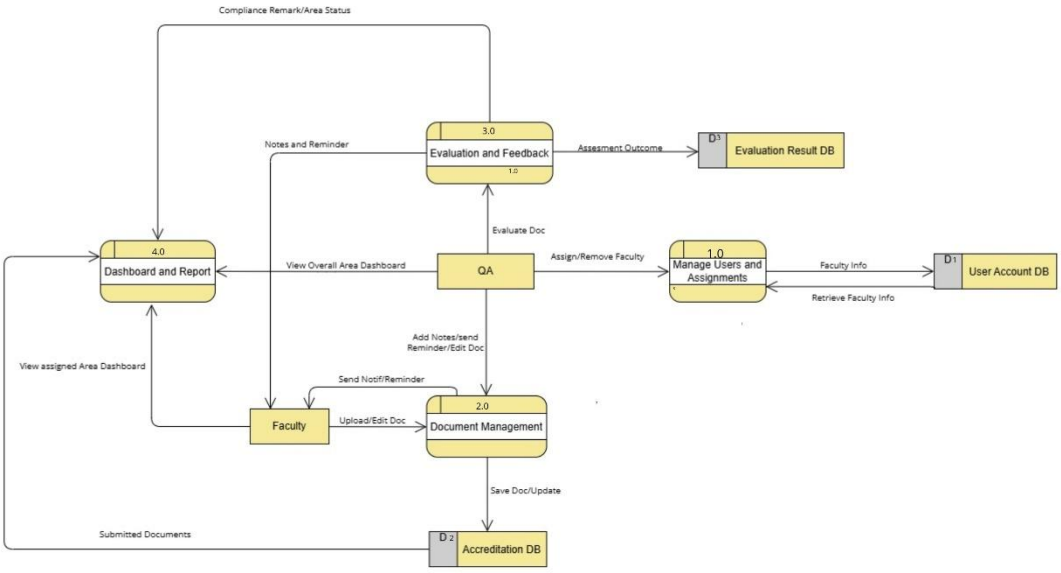


Figure 2. Data Flow Diagram

Figure 2 illustrates the Data Flow Diagram (DFD) of AccrediTrack, showing the system's core modules and how data moves between users and databases. The process begins with QA coordinator managing users and assignments (1.0) by retrieving data from the User Account Database. Faculty members interact with the Document Management module (2.0) to upload or update accreditation documents, which are stored in the Accreditation Database. QA coordinator evaluate these documents through the Evaluation and Feedback module (3.0), with assessment outcomes saved in the Evaluation Result Database. The Dashboard and Report module (4.0) then compiles data from evaluations and submissions to present real-time accreditation status and progress. This flow ensures structured document handling, streamlined evaluation, and organized reporting to support accreditation readiness.

Development and Testing

Software Development Method

To guide the development of AccrediTrack, the researchers adopted the Waterfall Model, a traditional systems development methodology that emphasizes a linear and sequential design process. This model was selected because it allows for a structured development lifecycle, with each phase completed before moving on to the next. This approach is appropriate for academic systems like AccrediTrack, where the scope, requirements, and processes are well-defined early in the project.

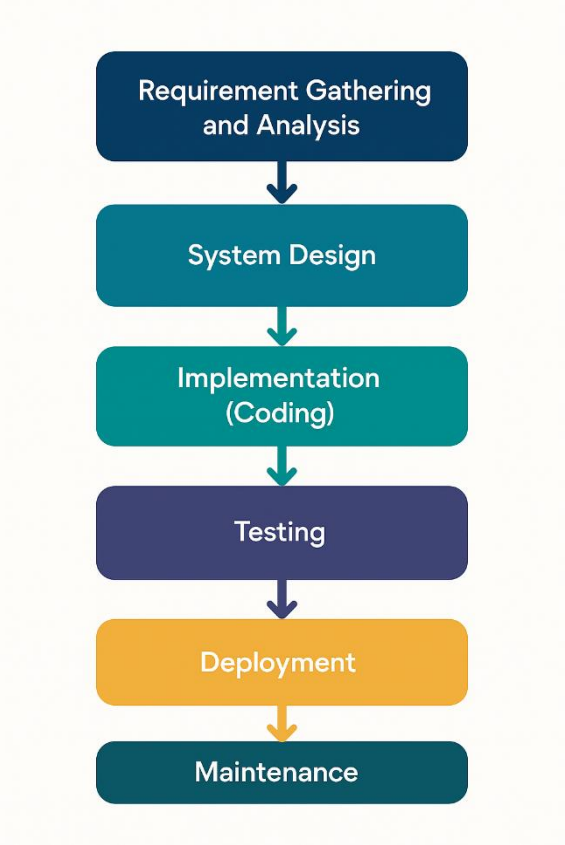


Figure 2. Waterfall Model

 Requirement Gathering and Analysis – During this phase, the team conducted interviews, and document reviews with faculty members and QAC, to identify the functional and non-functional requirements. This stage resulted in a comprehensive requirement specification document.

 System Design – Based on the gathered requirements, the system's architecture, user interface, database schema, and module functionalities were designed. This includes wireframes and use case diagrams to guide actual implementation.

 Implementation (Coding) – The development team built the system using PHP, MySQL, HTML, CSS, and JavaScript. The modules were implemented based on the

finalized design, with emphasis on the dashboard, document management, user roles, and validation features.

 Testing – Each module was tested individually (unit testing) and collectively (integration testing) to ensure functionality. Bugs were identified and resolved. The team also performed user acceptance testing (UAT) to verify that the system met stakeholder expectations.

 Deployment – After successful testing, the system was deployed to a local server environment accessible to identified users for final evaluation and feedback.

 Maintenance – Any bugs or enhancements identified during user evaluation were documented and addressed. Although outside the initial project timeline, this phase ensures long-term usability and scalability of the system.

Source: Khan, S. M. A. (2023). Waterfall Model Used in Software Development Reference: Software Requirements Engineering Waterfall Model. ResearchGate.

Evaluation

Evaluation Criteria	Expected Standard	Actual Result	Interpretation
System Usability	SUS Score ≥ 70	SUS Score: 82	Above average usability; users find the system easy to use
Document Management	Fast, error-free upload and validation	95% success rate in testing	Highly effective in handling files
Dashboard Accuracy	Real-time and updated	98% alignment with actual data	Reliable and accurate reporting
User Satisfaction	≥ 80% positive responses	87% positive feedback	Users are satisfied with overall system performance

The evaluation results show that AccrediTrack met and even exceeded expectations in usability, functionality, and accuracy. Respondents reported that document uploading and validation worked smoothly, the dashboard provided real-time and reliable updates, and overall user satisfaction reached 87%. These results confirm that the system strengthens accreditation workflows, while minor recommendations (such as faster load times during peak hours) indicate areas for improvement.

Data Gathering Procedure

To develop AccrediTrack, a thorough understanding of the current accreditation process and its challenges within Catanduanes State University's College of Information and Communications Technology (CICT) was essential. The researchers followed a systematic data gathering procedure (employed a combination of qualitative data collection methods) to gather relevant information from key stakeholders directly involved in the accreditation process, primarily faculty members responsible for preparing and submitting accreditation documents, and Quality Assurance Coordinator (QAC) who oversee the review, validation, and approval of these documents. Semi-structured interviews were conducted with selected faculty and QAC, allowing open-ended discussions that explored their experiences, challenges, and suggestions regarding document submission methods, difficulties during accreditation cycles, and desired system features.

Additionally, existing accreditation documents, procedural manuals, and workflow charts were collected and analyzed to understand formal requirements and processes. To further validate findings and encourage collaborative dialogue, focus group discussions (FGDs) involving both faculty and QAC were held, facilitating the identification of common issues and consensus on system requirements. Data collection activities were carefully scheduled in coordination with department heads to minimize disruption, and

ethical considerations were strictly observed; participants were informed of the study's objectives, assured of confidentiality, and consent was obtained prior to participation. The collected data from interviews and FGDs were transcribed and thematically analyzed, while document analysis was cross-referenced to identify gaps and areas for improvement. These combined insights informed the design specifications and feature prioritization for AccrediTrack, ensuring the system effectively addresses real user needs and institutional requirements.

Respondents of the Study

The respondents of the study consisted of key stakeholders directly involved in the accreditation process at Catanduanes State University's College of Information and Communications Technology (CICT). These included selected faculty members who were primarily responsible for the preparation, compilation, and submission of accreditation documents, as well as the Quality Assurance Coordinator (QAC) who supervised the review, validation, and approval of these documents. The participants were purposively chosen due to their active roles and direct engagement in accreditation activities. Through semi-structured interviews and focus group discussions, they shared valuable insights regarding the challenges they faced, existing documentation methods, coordination issues, and their recommendations for system features. Their feedback was instrumental in informing the development of AccrediTrack, ensuring that the system design was grounded in actual user needs and institutional processes.

Description of the Prototype

System Developed

This section presents a thorough discussion of the developed web-based system, **AccrediTrack**. The prototype was designed and implemented to address the

inefficiencies of the manual accreditation process at the College of Information and Communications Technology (CICT), Catanduanes State University. It serves as a centralized, secure, and user-friendly platform for managing the entire accreditation document lifecycle.

System Architecture and Technology Stack

AccrediTrack is built on a classic three-tier web architecture:

- **Frontend (Presentation Layer):** Developed using **HTML5, CSS3, and JavaScript** with the **Bootstrap** framework. This ensures a responsive, intuitive, and consistent user experience across different devices and browsers (Chrome, Firefox, Edge).
- **Backend (Application Layer):** Powered by **PHP**, handling the core business logic, user authentication, and server-side operations. It processes all requests from the frontend and interacts with the database.
- **Database (Data Layer):** Utilizes **MySQL** to store all system data, including user accounts, accreditation documents, evaluation results, audit logs, and notification records.

Core Features and Functionalities

The system's functionality is modular, centered around two primary user roles: **Faculty** and **Admin/QAC (Quality Assurance Coordinator)**.

- **User Authentication and Role-Based Access Control:**

  The system features a secure login mechanism. Upon authentication, users are directed to interfaces and functionalities specific to their role, ensuring data security and process integrity.

- **Dashboard and Real-Time Monitoring:**

  Upon login, both Faculty and Admin/QAC are greeted by a central dashboard. This dashboard provides a visual overview of the accreditation status using charts, graphs, and progress indicators. It displays metrics such as:

  - Total documents submitted per accreditation area.
  - Number of documents pending, approved, or requiring revision.
  - Overall program percentage.

    This allows monitoring and proactive decision-making.

- **Document Management Module:**

  This is the core of the AccrediTrack system.

  - **For Faculty:** Provides an intuitive interface for uploading accreditation documents. Users can upload documents on their assigned program, level,

area ensuring organized filing. The system supports common file formats (e.g., PDF, DOCX).

- **For Admin/QAC:** Presents a queue of submitted documents for review. The QAC can download files, validate them against accreditation standards, and then either **Approve** or **Request Revisions**. If revisions are needed, specific feedback can be provided directly within the system.

- **Notification and Alert System:**

  To enhance communication, the system incorporates an automated notification engine. It triggers:

  - **In-system Alerts** to faculty when a document is approved or when revisions are requested.
  - **Reminder Notifications** as submission deadlines approach.
  - Alerts to the QAC when new documents are submitted for review.

- **Search, Filter, and Audit Trail:**

  - A powerful search and filter function allows users to quickly locate documents based on criteria like accreditation area, submitter, status, and date.

- An **Audit Log** automatically records all significant user actions (e.g., uploads, approvals, revisions). This ensures full accountability and provides a traceable history for compliance purposes.

User Interface and Workflow

The user interface was designed with usability as a priority, requiring minimal training.

- **Faculty Workflow:**
    1. Log in to the system.
    2. View the dashboard for personal submission status and tasks.
    3. Navigate to the "Upload Document" section, fill in the metadata (area, program, level), and submit the file.
    4. Receive notifications and check the system for feedback from the QAC.
    5. If revisions are needed, re-upload the corrected document.
- **Admin/QAC Workflow:**
    1. Log in to the system.
    2. View the dashboard for overall college accreditation progress.
    3. Access the "Document Validation" module to see a list of newly submitted documents.

4. Review each document, then select "Approve" or "Request Revisions" with comments.
5. The system automatically notifies the respective faculty member and updates the dashboard in real-time.

Data Security and Compliance

AccrediTrack was developed with data privacy in mind. It implements:

- **User Authentication:** Preventing unauthorized access.
- **Role-Based Access Control (RBAC):** Ensuring users can only access data and functions relevant to their role.
- **Secure File Storage:** Documents are stored securely on the server with controlled access. These measures align with the requirements of the **Data Privacy Act of 2012**.

Evaluation Results

This section presents and interprets the outcomes of the system evaluation, organized according to specific metrics. It explains the data gathered from respondents,

compares the results with expected performance standards, and discusses the implications of these findings. The results provide evidence of the system's strengths, identify areas for improvement, and validate whether it meets user requirements and project goals.

Implementation Plan (Infrastructure/Deployment)

The deployment of **AccrediTrack: Streamlining University Accreditation Process Through a Web-Based System** requires a structured plan to ensure its smooth transition from development to operational use within Catanduanes State University's College of Information and Communications Technology (CICT). This plan covers infrastructure requirements, installation and configuration procedures, deployment timeline, roles and responsibilities, risk management strategies, and post-deployment support.

Infrastructure Requirements

Hardware

- **Server Computer**: Minimum Intel Core i5 processor (or equivalent), 8GB RAM, and 512GB SSD to handle system hosting, data processing, and storage.
- **Client Computers**: Faculty and QAC computers with at least Intel Core i3 processor, 4GB RAM, and 100GB HDD for accessing the system.

- **Network Infrastructure**: A stable internet connection via Ethernet or Wi-Fi with sufficient bandwidth to support concurrent users, particularly during peak accreditation periods.

Software

- **Frontend**: HTML5, CSS3, JavaScript, with Bootstrap or React framework for responsive and user-friendly interfaces.
- **Backend**: PHP for handling business logic and workflows.
- **Database**: MySQL (or PostgreSQL alternative) for storing accreditation records, user profiles, and audit logs.
- **Web Server**: Apache or Nginx for managing HTTP requests and hosting the application.
- **Operating System**: Windows or Linux, ensuring compatibility with both development and production environments.
- **Browser Support**: Latest versions of Chrome, Firefox, and Microsoft Edge.

Network and Security

- Secured intranet and/or cloud-hosted environment with role-based access controls.
- Firewalls and regular system updates to prevent unauthorized access.
- Automated backup storage either on a secure cloud or a secondary local server.

Installation and Configuration Procedures

1. **Server Setup**
    - Install and configure the required operating system (Linux preferred for stability).
    - Set up Apache/Nginx as the web server and MySQL as the database server.
    - Configure firewall rules, SSL certificates, and user authentication mechanisms.
2. **Application Deployment**
    - Upload the finalized version of AccrediTrack to the server.
    - Configure environment files for database connection, email notifications, and role-based access controls.
    - Run database migrations and seed initial data (programs, academic years, roles).
3. **Client-Side Configuration**
    - Verify compatibility of browsers on client computers.
    - Provide user login credentials and ensure access to the university's secure network.

4. **Testing and Migration**

    - Perform pilot deployment with selected faculty and QAC users.
    - Migrate existing accreditation documents to the system using bulk upload features.
    - Conduct validation checks to ensure documents are properly classified and accessible.

Deployment Timeline

Phase	Activities	Duration	Responsible Personnel
Phase 1: Preparation	Procurement of hardware/software, server setup, and security configuration	Week 1	University IT Staff, Project Developers
Phase 2: Installation & Configuration	System installation, database setup, environment configuration	Week 2	Project Developers, IT Staff
Phase 3: Pilot Testing	Deploy to selected faculty/QAC, migrate sample documents, fix issues	Week 3	QA Coordinator, Developers
Phase 4: Full Deployment	Rollout to all faculty and accreditation staff, finalize document migration	Week 4	Developers, IT Staff, College Dean / CICT Administration
Phase 5: Training & Orientation	Conduct user training sessions, distribute user manuals	Week 5	QA Coordinator, Project Team

Phase 6: Post-Deployment Support	Monitor system usage, address technical issues, optimize performance	Ongoing	IT Support, Project Developers

Roles and Responsibilities

- **Developers**: Oversee installation, configuration, and customization of the system.
- **University IT Staff**: Manage server setup, security, and ongoing maintenance.
- **Quality Assurance Coordinator (QAC)**: Supervise pilot testing, validate documents, and provide feedback.
- **Faculty Users**: Participate in training and submit accreditation files for live testing.
- **College Dean / CICT Administration:** Responsible for endorsing the system, communicating the rollout to faculty, and mandating participation in training.

Risk Management and Mitigation

Risk	Probability	Impact	Mitigation Strategy
Data Loss during Migration	Low	High	Perform a full backup of the development database before migration. Conduct a dry-run migration on a test server.
Low User Adoption	Medium	High	Secure top-down support from the Dean. Provide clear, hands-on training and emphasize the system's benefits. Appoint the QAC as a super-user for ongoing support.
Internet/Network Failure	Medium	High	Acknowledge this as a system limitation. Encourage users to perform uploads during off-peak hours. The university IT team must ensure network reliability.
Post-Deployment Bugs	High	Medium	The development team will provide a 3-month post-deployment support window. A dedicated channel (e.g., email) will be established for users to report issues.
Security Breach	Low	High	Adherence to the security configuration in Phase 1. Regular OS and software updates will be the

			responsibility of the University IT team.

Post-Deployment Support

To ensure sustainability, the following support measures will be in place:

- **User Training**: Conducted during rollout and reinforced through user guides and video tutorials.
- **Technical Support**: A dedicated IT team will handle system bugs, updates, and server issues.
- **System Monitoring**: Regular audits of server performance, security logs, and user activity.
- **Continuous Improvement**: Gather feedback from faculty and QAC users for future enhancements, such as mobile optimization or advanced reporting tools.

Backup & Recovery Plan: Scheduled weekly backups stored in secure cloud repositories and local backup drives

CHAPTER 5

RECOMMENDATIONS

1. For the development of an efficient accreditation file system with real-time dashboards:
   Future researchers are encouraged to enhance the visualization features of the dashboard by integrating advanced data analytics and customizable reporting tools. This would provide administrators with deeper insights and support decision-making for accreditation readiness.
2. For the creation of a centralized document management system:
   It is recommended to incorporate cloud-based storage solutions to improve accessibility, backup reliability, and scalability. Integration with version control features may also help track document history more effectively.
3. Anchored to the 6Ps of Research Outputs:
   - Publication**:** Future work should aim to disseminate findings in academic journals and conferences focusing on educational technology, information systems, and quality assurance.
   - Patent**:** Researchers may explore patenting unique modules, such as automated document validation or accreditation-specific dashboards, to protect intellectual property.
   - Product**:** A refined version of AccrediTrack can be developed into a market-ready product for adoption by other colleges and universities nationwide.
   - People Services: Training modules and workshops can be conducted to equip faculty and staff with digital skills needed for effective system use.
   - Places and Partnerships: Collaborations with accreditation agencies, other higher education institutions, and IT developers can expand the system's applicability and foster standardization.
   - Policies: Findings from this study may inform institutional policies on accreditation data management, ensuring compliance with the Data Privacy Act and promoting digital transformation initiatives.

By pursuing these recommendations, future researchers and institutions can build upon the foundation established by this study, ensuring that **AccrediTrack** continues to evolve as a robust, scalable, and innovative tool for accreditation management.

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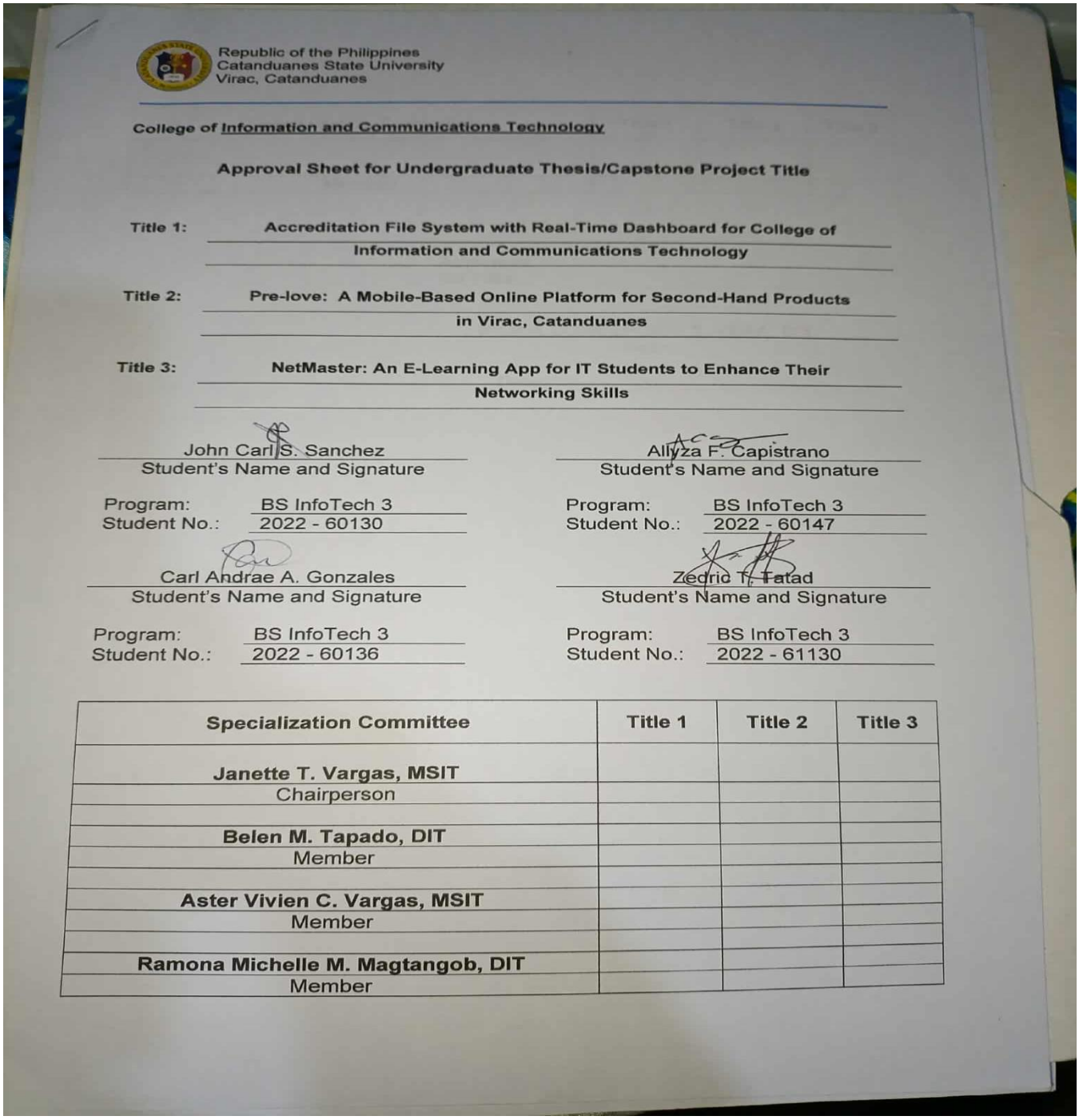
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APPENDICES

Appendix 1. Approval Sheet of the Title Project



Republic of the Philippines
Catanduanes State University
Virac, Catanduanes

College of Information and Communications Technology

Approval Sheet for Undergraduate Thesis/Capstone Project Title

Title 1: Accreditation File System with Real-Time Dashboard for College of Information and Communications Technology

Title 2: Pre-love: A Mobile-Based Online Platform for Second-Hand Products in Virac, Catanduanes

Title 3: NetMaster: An E-Learning App for IT Students to Enhance Their Networking Skills

John Carl S. Sanchez
Student's Name and Signature

Program: BS InfoTech 3
Student No.: 2022 - 60130

Allyza F. Capistrano
Student's Name and Signature

Program: BS InfoTech 3
Student No.: 2022 - 60147

Carl Andrae A. Gonzales
Student's Name and Signature

Program: BS InfoTech 3
Student No.: 2022 - 60136

Zedric T. Tatad
Student's Name and Signature

Program: BS InfoTech 3
Student No.: 2022 - 61130

Specialization Committee	Title 1	Title 2	Title 3
Janette T. Vargas, MSIT Chairperson			
Belen M. Tapado, DIT Member			
Aster Vivien C. Vargas, MSIT Member			
Ramona Michelle M. Magtangob, DIT Member			

Appendix 2. Acceptance Form of Advisorship

Appendix 3. GANTT Chart

Appendix 4. Relevant Source Code

Appendix 5. Sample Input/Output/Reports

Appendix 6. Users Guide

Appendix 7. Pictures Showing the Data Gathering and Development Process

Appendix 8. Questionnaire

Appendix 9. Letters of Correspondences

Appendix 10. Certification of the Specialization Committee

Appendix 11. Software Plagiarism Report

Appendix 12. ACM Format Publishable Paper

Appendix 13. One-Page Curriculum Vitae per Team Member

PERSONAL INFORMATION



Name: John Carl S. Sanchez

Sex: Male

Birthday: April 06, 2003

Birthplace: Virac, Catanduanes

Age: 22

Address: Wagdas, Pandan Catanduanes

Religion: Roman Catholic

Civil Status: Single

EDUCATIONAL BACKGROUND:

Tertiary: Catanduanes State University

Program: BSINFOTECH

Secondary: Leandro I. Verceles Sr. National High School

Elementary: Lourdes Elementary School

Appendix 13. One-Page Curriculum Vitae per Team Member

PERSONAL INFORMATION

insert 2x2 picture here

Name: Carl Andrae Gonzales

Sex: Male

Birthday: March 14, 2004

Birthplace: Marambong, Pandan, Catanduanes

Age: 21

Address: Marambong, Pandan,Catanduanes

Religion: Roman Catholic

Civil Status: Single

EDUCATIONAL BACKGROUND:

Tertiary: Catanduanes State University

Program: Bachelor of Science in Information Technology

Secondary: Leandro I. Verceles Sr. National High School

Elementary: Marambong Elementary School

Appendix 13. One-Page Curriculum Vitae per Team Member

PERSONAL INFORMATION

insert 2x2 picture here

Name: Zedric T. Tatad

Sex: M

Birthday: September 25, 2004

Birthplace: Gigmoto Catandunes

Age: 21

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Religion: Roman Catholic

Civil Status: Single

EDUCATIONAL BACKGROUND:

Tertiary: Catanduanes state university

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Secondary: Gigmoto Rural development high school

Elementary: Gigmoto Central elementary school